

The $U(3)$ Gelfand–Tsetlin System: Exact Fibers, Linear Dynamics, and Unshifted Quantization

HCS-C365 theorem package

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Abstract

We close the classical and quantum Gelfand–Tsetlin descriptions for a Hermitian $U(3)$ coadjoint orbit under explicit conventions. An arrow completion proves the exact interlacing image; nested-group flows give every regular Lagrangian torus; character duality classifies every linear Hamiltonian closure and least period. With integral highest weight and no half-form correction, branching gives the complete basis with unshifted GT labels and the Weyl count. Repeated weights, facets, and nonintegral spectra are separated sharply. Exact enumeration is a regression receipt, not the proof.

1 Frozen orbit and exact theorem

Let $\lambda_1 \geq \lambda_2 \geq \lambda_3$, and let $\mathcal{O}_\lambda$ be the orbit of Hermitian matrices with this spectrum. For skew-Hermitian X, Y , freeze

$$\omega_A([X, A], [Y, A]) = -i \operatorname{Tr}(A[X, Y]), \quad f_X(A) = -i \operatorname{Tr}(AX), \quad \iota_{X_{f_X}} \omega = df_X.$$

Thus $X_{f_X}(A) = [X, A]$. Write $B = A^{(2)}$ for the northwest 2×2 minor, let $y_1 \geq y_2$ be its eigenvalues, and put $x = A_{11}$.

Theorem 1 *The map $F = (y_1, y_2, x)$ has image*

$$\Delta_\lambda = \{\lambda_1 \geq y_1 \geq \lambda_2 \geq y_2 \geq \lambda_3, \quad y_1 \geq x \geq y_2\}.$$

If $\lambda_1 > \lambda_2 > \lambda_3$, every strict-interlacing fiber is exactly an effective Lagrangian T^3 -orbit, with each circle normalized to period 2π .

For $H_\nu = \nu_1 y_1 + \nu_2 y_2 + \nu_3 x$, its regular flow is $\theta(t) = \theta_0 + t\nu \pmod{2\pi}$. Let K_ν be the subgroup annihilated by every $k \in \mathbb{Z}^3$ satisfying $k \cdot \nu = 0$. The orbit closure is the coset $\theta_0 + K_\nu$, and $\theta_0 + K_\nu = K_\nu$ exactly when $\theta_0 \in K_\nu$, with $\dim K_\nu = \operatorname{rank}_{\mathbb{Q}}\{\nu_1, \nu_2, \nu_3\}$. It is fixed for $\nu = 0$. For $\nu \neq 0$, it is periodic precisely when $\nu = \alpha m$, where $\alpha > 0$ and $m \in \mathbb{Z}^3$ is primitive; its least period is $2\pi/\alpha$.

If $\lambda \in \mathbb{Z}^3$ is dominant, take prequantum curvature $-i\omega$ and no half-form correction. Borel–Weil quantization has a multiplicity-free GT branching basis indexed, without a ρ -shift, by

$$\lambda_1 \geq m_{21} \geq \lambda_2 \geq m_{22} \geq \lambda_3, \quad m_{21} \geq m_{11} \geq m_{22},$$

with joint branching labels (m_{21}, m_{22}, m_{11}) , and

$$\dim Q(\mathcal{O}_\lambda) = \frac{(\lambda_1 - \lambda_2 + 1)(\lambda_2 - \lambda_3 + 1)(\lambda_1 - \lambda_3 + 2)}{2}.$$

2 Interlacing and constructive surjectivity

Cauchy interlacing gives all six inequalities. Conversely, at any point of Δ_λ , set

$$B_{11} = x, \quad B_{12} = B_{21} = r, \quad B_{22} = y_1 + y_2 - x, \quad r^2 = (y_1 - x)(x - y_2).$$

This has eigenvalues y_1, y_2 . In a basis diagonalizing B , consider

$$A'_{11} = y_1, \quad A'_{22} = y_2, \quad A'_{12} = 0, \quad A'_{13} = z_1, \quad A'_{23} = z_2, \quad A'_{33} = c = \lambda_1 + \lambda_2 + \lambda_3 - y_1 - y_2,$$

with the lower-triangular entries fixed by Hermitian symmetry. If $P(t) = \prod_{j=1}^3 (t - \lambda_j)$, choose

$$|z_1|^2 = -\frac{P(y_1)}{y_1 - y_2}, \quad |z_2|^2 = \frac{P(y_2)}{y_1 - y_2}.$$

Interlacing makes both quantities nonnegative. The monic cubic $\det(tI - A')$ has the correct trace and agrees with P at y_1, y_2 , hence equals P . Conjugating the first two coordinates back retains $A_{11} = x$. Endpoint coincidences follow by setting the vanishing arrow coordinate to zero, or by continuity. This proves the exact image.
round zero interlacing and arrow completion closure